

Intel Core Ultra 7 270K Plus

Punching above its weight

The Intel Core Ultra 7 270K Plus might have come out of the blue, but it feels like the desktop Arrow Lake part that should have arrived first. It's not exactly the clean-sheet rethink of Intel's mainstream performance formula but it certainly does come across as a more assertive version of it. The Core Ultra 7 270K Plus brings the core count up to 24 with an 8-Performance plus 16-Efficiency core layout, keeps boost clocks high, and leans on platform refinements such as faster die-to-die communication and Intel's new Binary Optimization Tool to shore up gaming and responsiveness. The 270K Plus is competitive in productivity, and convincing than the first wave of desktop Core Ultra chips, if it still does not flatten AMD's best X3D parts in every gaming scenario.

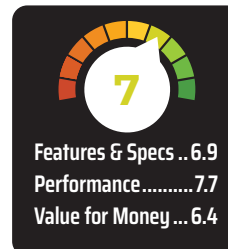
TEST RIG

The review unit was tested on Gigabyte's Z890 AORUS PRO ICE motherboard with 64GB of G.Skill Trident Z5 memory running at 7200 MT/s in a 2x 32 GB configuration. Storage duties were handled by a Kingston Renegade PCIe Gen 4.0 SSD, while cooling came from the Corsair iCUE Link Titan X360 RX LCD, which is a sensible pairing for a desktop processor that can pull up to 250W at full tilt. Power was supplied by the Cooler Master MWE Gold 850 V2. It is a well-balanced, modern test platform that gives the 270K Plus the kind of memory bandwidth and thermal headroom it needs to show what it can do without obvious platform bottlenecks getting in the way. Here's the full specs:

- **CPU:** Intel Core Ultra 7 270K Plus
- **Motherboard:** Z890 AORUS PRO ICE



- **RAM:** 2x G.Skill Trident Z5 32 GB 7200 MT/s
- **SSD:** Kingston Renegade PCIe Gen 4.0 SSD
- **Cooler:** Corsair iCUE Link Titan X360 RX
- **PSU:** Cooler Master MWE Gold 850 V2



PERFORMANCE

Cinebench 2024

Cinebench 2024 is based on Maxon's Cinema 4D software, designed to evaluate a processor's performance in rendering complex 3D scenes. It tests both single-core and multi-core capabilities, highlighting how efficiently a CPU handles multi-threaded tasks common in professional rendering workflows. The benchmark utilizes modern instruction sets and large datasets, providing an up-to-date assessment of CPU performance in contemporary applications.

WinRAR

WinRAR Benchmark evaluates a processor's performance in data compression

and decompression tasks, common in file archiving and management. It tests single-threaded performance, memory bandwidth, and latency, as these factors influence compression speed. While WinRAR isn't fully optimized for multi-core CPUs, processors with higher Instructions Per Cycle (IPC) and clock speeds perform better. For more modern algorithms, we use 7-Zip.

AIDA64

AIDA64 Memory Benchmark measures the memory bandwidth and latency of a system, highlighting how quickly data transfers between the CPU and RAM. It tests read, write, and copy speeds, providing insights into the efficiency of the memory subsystem, including RAM speed, timings, and memory controller performance. AIDA64 also has benchmarks to evaluate how well a processor can handle AES, ZLib and SHA3 encryption and decryption tasks.

y-cruncher

The y-cruncher benchmark computes mathematical constants like Pi to a high number of digits—in this case, 2.5 billion—testing a processor's multi-threaded performance and memory subsystem under heavy computational load. It stresses the CPU's integer and floating-point units, cache hierarchy, and memory bandwidth. It's particularly useful for evaluating system stability under sustained heavy workloads, making it valuable for users who are overclocking their systems.

Procyon AI Computer Vision

The UL Procyon AI Computer Vision benchmark measures the performance of AI inference engines to understand how well processors can handle machine-vision tasks using popular neural networks.

3DMark Time Spy (iGPU)

3DMark Time Spy is a DirectX 12 benchmark evaluating graphics performance.